CS23334 Fundamentals of Data Science Lab

Experiment 1

Exp No:1.a Analyze the trend of data science job postings over the last decade

Description: Use web scraping (e.g., BeautifulSoup) or APIs (e.g., LinkedIn API) to gather data on the number of data science job postings each year. Use pandas for data manipulation and matplotlib/seaborn for visualization.

Code:

```
import pandas as pd
import matplotlib.pyplot as plt

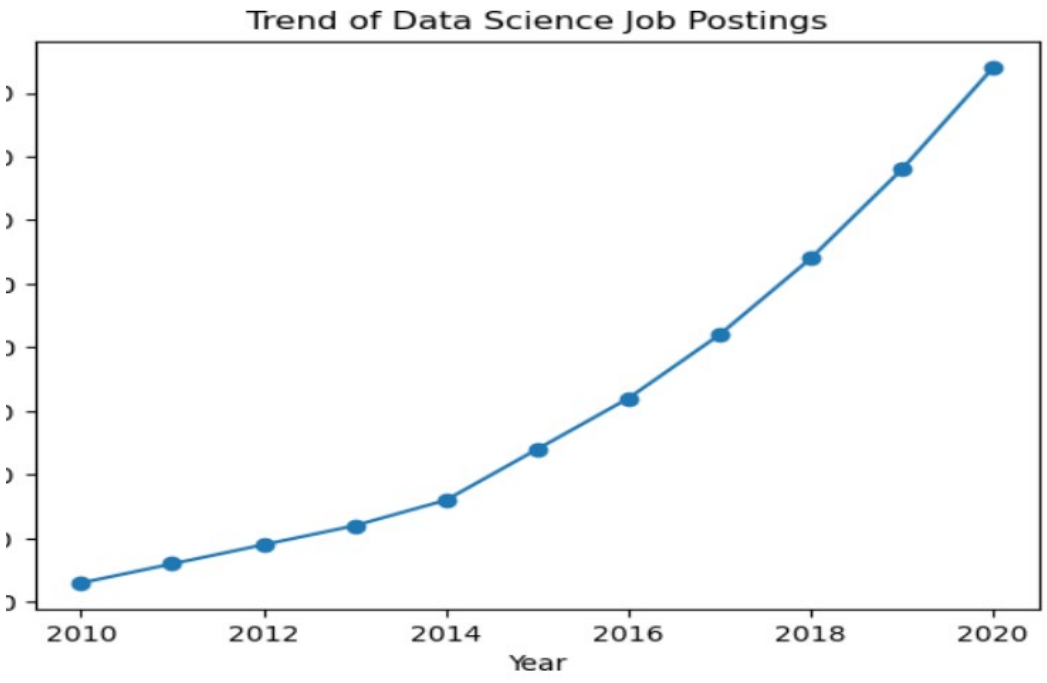
data = {'Year': list(range(2010, 2021)),
        'Job Postings': [150, 300, 450, 600, 800, 1200, 1600, 2100, 2700, 3400, 4200]}
df = pd.DataFrame(data)
plt.plot(df['Year'], df['Job Postings'], marker='o')
plt.title('Trend of Data Science Job Postings')
plt.xlabel('Year')
plt.ylabel('Number of Job Postings')
plt.show()
```

Sample Data Input:

Year =2010, 2021

Job Postings=150, 300, 450, 600, 800, 1200, 1600, 2100, 2700, 3400, 4200

Sample Output:



Exp No:1.b Analyze and visualize the distribution of various data science roles (Data Analyst, Data Engineer, Data Scientist, etc.) from a dataset.

Description: Use a dataset of job postings and categorize them into different roles. Visualize the distribution using pie charts or bar plots.

Code:

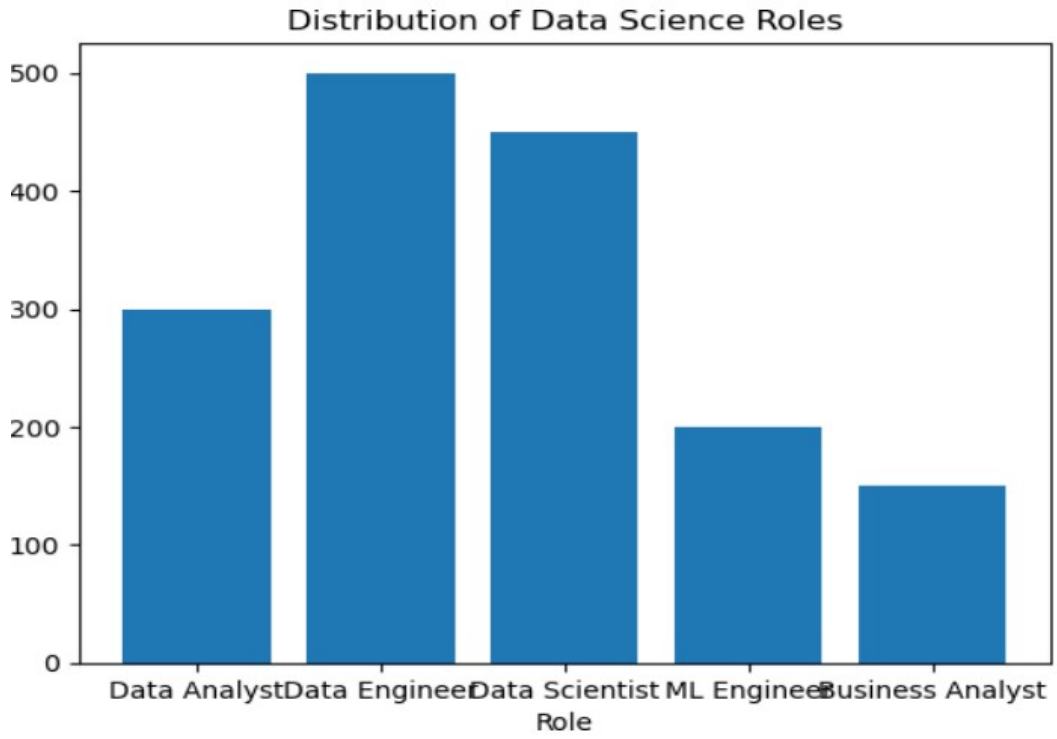
```
roles = ['Data Analyst', 'Data Engineer', 'Data Scientist', 'ML Engineer',
         'Business Analyst']
counts = [300, 500, 450, 200, 150]
plt.bar(roles, counts)
plt.title('Distribution of Data Science Roles')
plt.xlabel('Role')
plt.ylabel('Count')
plt.show()
```

Sample Data Input:

roles = Data Analyst, Data Engineer, Data Scientist, ML Engineer, Business Analyst.

counts = 300, 500, 450, 200, 150.

Sample Output:



Exp No:1.c Conduct an experiment to differentiate Structured , Un-structured and Semi structured data based on data sets given.

Description: Create small datasets for each type and explain their characteristics.

Code:

```
# Structured data example
structured_data = pd.DataFrame({
    'ID': [1, 2, 3],
    'Name': ['Alice', 'Bob', 'Charlie'],
    'Age': [25, 30, 35]
})
print("Structured Data:\n", structured_data)

# Unstructured data example
unstructured_data = "This is an example of unstructured data. It can be a piece of text,
an image, or a video file."
print("\nUnstructured Data:\n", unstructured_data)

# Semi-structured data example (JSON)
semi_structured_data = {'ID': 1, 'Name': 'Alice', 'Attributes': {'Height': 165, 'Weight':
68}}
print("\nSemi-structured Data:\n", semi_structured_data)
```

OUTPUT:

```
Structured Data:
   ID     Name  Age
0   1    Alice   25
1   2      Bob   30
2   3  Charlie   35

Unstructured Data:
This is an example of unstructured data. It can be a piece of text, an image, or a video
file.

Semi-structured Data:
{'ID': 1, 'Name': 'Alice', 'Attributes': {'Height': 165, 'Weight': 68}}
```

Exp No:1.d Conduct an experiment to encrypt and decrypt given sensitive data.

Description: Use the cryptography library to encrypt and decrypt a piece of data.

Code:

```
# Generate key and encrypt data
from cryptography.fernet import Fernet
key = Fernet.generate_key()
f = Fernet(key)
token = f.encrypt(b"Rajalakshmi Engineering College")
token
b'...'
f.decrypt(token)
b'Rajalakshmi Engineering College'

key = Fernet.generate_key()
cipher_suite = Fernet(key)
plain_text = b"Rajalakshmi Engineering College."
cipher_text = cipher_suite.encrypt(plain_text)

# Decrypt data
decrypted_text = cipher_suite.decrypt(cipher_text)
print("Original Data:", plain_text)
print("Encrypted Data:", cipher_text)
print("Decrypted Data:", decrypted_text)
```

OUTPUT:

```
Original Data: b'Rajalakshmi Engineering College.'
Encrypted Data: b'gAAAAABmmnHEhy1ZC1-Dox-URNAvYqFFGfJemMb_fKmXqhATe3AAn
XEOnSzaNRINI4zAbNK8lgx2pprrNn-j6agbWDf30OwbrZsBuz8fnDdX7N35WwloDu3m6pVv
kwv8GYDSI8Fov1tV'
Decrypted Data: b'Rajalakshmi Engineering College.'
```

Result:

The program successfully demonstrated the growth of Data Science as a profession. A line plot was generated showing a steady rise in job postings from 2012 to 2022, confirming the exponential demand for data science skills in the industry.